

$$\mathbf{G}_\theta \left(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t) \right)$$

$$\begin{aligned} & \tilde{w}(s, t) \left\| \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r') \right\|_2 \\ &= \tilde{w}(s, t) \left\| c_{\text{skip}}(s, t) \mathbf{x}_t + c_{\text{out}}(s, t) \hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) - \left(c_{\text{skip}}(s, r) \mathbf{x}_r' + c_{\text{out}}(s, r) \hat{\mathbf{G}}(\mathbf{x}_r', s, r) \right) \right\|_2 \\ &= \tilde{w}(s, t) c_{\text{out}}(s, t) \left\| \hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) - \frac{1}{c_{\text{out}}(s, t)} \left(c_{\text{skip}}(s, r) \mathbf{x}_r' + c_{\text{out}}(s, r) \hat{\mathbf{G}}(\mathbf{x}_r', s, r) - c_{\text{skip}}(s, t) \mathbf{x}_t \right) \right\|_2 \end{aligned}$$

We simply set $\tilde{w}(s, t) = 1/c_{\text{out}}(s, t)$ for the overall weighting to be 1. This allows invariance of magnitude of kernels w.r.t. t .

C.9. Weighting Function $w(s, t)$

To review VDM (Kingma et al., 2021), the negative ELBO loss for diffusion model is

$$\mathcal{L}_{\text{ELBO}}(\theta) = \frac{1}{2} \mathbb{E}_{\mathbf{x}, \epsilon, t} \left[\left(-\frac{d}{dt} \lambda_t \right) \left\| \epsilon_\theta(\mathbf{x}_t, t) - \epsilon \right\|^2 \right] \quad (69)$$

where ϵ_θ is the noise-prediction network and $\lambda_t = \log\text{-SNR}_t$. The weighted-ELBO loss proposed in Kingma & Gao (2024) introduces an additional weighting function $w(t)$ monotonically increasing in t (monotonically decreasing in $\log\text{-SNR}_t$) understood as a form of data augmentation. Specifically, they use sigmoid as the function such that the weighted ELBO is written as

$$\mathcal{L}_{w\text{-ELBO}}(\theta) = \frac{1}{2} \mathbb{E}_{\mathbf{x}, \epsilon, t} \left[\sigma(b - \lambda_t) \left(-\frac{d}{dt} \lambda_t \right) \left\| \epsilon_\theta(\mathbf{x}_t, t) - \epsilon \right\|^2 \right] \quad (70)$$

where $\sigma(\cdot)$ is sigmoid function.

The α_t is tailored towards the Simple-EDM and Euler-FM parameterization as we have shown in Appendix C.5 that the networks $\sigma_d \mathbf{G}_\theta$ amounts to v -prediction in cosine and OT-FM schedules. Notice that ELBO diffusion loss matches ϵ instead of $\mathbf{v}$. Inspecting the gradient of Laplace kernel, we have (again, for simplicity we let $\hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) = \mathbf{G}_\theta(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t))$)

$$\begin{aligned} & \frac{\partial}{\partial \theta} e^{-\tilde{w}(s, t)} \left\| \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r') \right\|_2 / D \\ &= -\frac{\tilde{w}(s, t)}{D} e^{-\tilde{w}(s, t)} \left\| \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r') \right\|_2 / D \frac{\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r')}{\left\| \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r') \right\|_2} \frac{\partial}{\partial \theta} \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) \\ &= -\frac{\tilde{w}(s, t)}{D} e^{-\tilde{w}(s, t)} \left\| \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r') \right\|_2 / D \frac{\hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) - \hat{\mathbf{G}}_{\text{target}}}{\left\| \hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) - \hat{\mathbf{G}}_{\text{target}} \right\|_2} \frac{\partial}{\partial \theta} \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) \end{aligned}$$

for some constant $\hat{\mathbf{G}}_{\text{target}}$. We can see that gradient $\frac{\partial}{\partial \theta} \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$ is guided by vector $\hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) - \hat{\mathbf{G}}_{\text{target}}$. Assuming $\hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t)$ is v -prediction, as is the case for Simple-EDM parameterization with cosine schedule and Euler-FM parameterization with OT-FM schedule, we can reparameterize v - to ϵ -prediction with ϵ_θ as the new parameterization. We omit arguments to network for simplicity.

We show below that for both cases $\epsilon_\theta - \epsilon_{\text{target}} = \alpha_t(\hat{\mathbf{G}}_\theta - \hat{\mathbf{G}}_{\text{target}})$ for some constants ϵ_{target} and $\hat{\mathbf{G}}_{\text{target}}$. For Simple-EDM, we know x -prediction from v -prediction parameterization (Salimans & Ho, 2022), $\mathbf{x}_\theta = \alpha_t \mathbf{x}_t - \sigma_t \hat{\mathbf{G}}_\theta$, and we also know

x -prediction from ϵ -prediction, $\mathbf{x}_\theta = (\mathbf{x}_t - \sigma_t \epsilon_\theta) / \alpha_t$. We have

$$\frac{\mathbf{x}_t - \sigma_t \epsilon_\theta}{\alpha_t} = \alpha_t \mathbf{x}_t - \sigma_t \hat{\mathbf{G}}_\theta \quad (71)$$

$$\iff \mathbf{x}_t - \sigma_t \epsilon_\theta = \alpha_t^2 \mathbf{x}_t - \alpha_t \sigma_t \hat{\mathbf{G}}_\theta \quad (72)$$

$$\iff (1 - \alpha_t^2) \mathbf{x}_t + \alpha_t \sigma_t \hat{\mathbf{G}}_\theta = \sigma_t \epsilon_\theta \quad (73)$$

$$\iff \sigma_t^2 \mathbf{x}_t + \alpha_t \sigma_t \hat{\mathbf{G}}_\theta = \sigma_t \epsilon_\theta \quad (74)$$

$$\iff \epsilon_\theta = \sigma_t \mathbf{x}_t + \alpha_t \hat{\mathbf{G}}_\theta \quad (75)$$

$$\iff \epsilon_\theta - \epsilon_{\text{target}} = \sigma_t \mathbf{x}_t + \alpha_t \hat{\mathbf{G}}_\theta - (\sigma_t \mathbf{x}_t + \alpha_t \hat{\mathbf{G}}_{\text{target}}) \quad (76)$$

$$\iff \epsilon_\theta - \epsilon_{\text{target}} = \alpha_t (\hat{\mathbf{G}}_\theta - \hat{\mathbf{G}}_{\text{target}}) \quad (77)$$

For Euler-FM, we know x -prediction from v -prediction parameterization, $\mathbf{x}_\theta = \mathbf{x}_t - t \hat{\mathbf{G}}_\theta$ and we also know x -prediction from ϵ -prediction, $\mathbf{x}_\theta = (\mathbf{x}_t - t \epsilon_\theta) / (1 - t)$. We have

$$\frac{\mathbf{x}_t - t \epsilon_\theta}{1 - t} = \mathbf{x}_t - t \hat{\mathbf{G}}_\theta \quad (78)$$

$$\iff \mathbf{x}_t - t \epsilon_\theta = (1 - t) \mathbf{x}_t - t(1 - t) \hat{\mathbf{G}}_\theta \quad (79)$$

$$\iff t \mathbf{x}_t + t(1 - t) \hat{\mathbf{G}}_\theta = t \epsilon_\theta \quad (80)$$

$$\iff \epsilon_\theta = \mathbf{x}_t + (1 - t) \hat{\mathbf{G}}_\theta \quad (81)$$

$$\iff \epsilon_\theta = \mathbf{x}_t + \alpha_t \hat{\mathbf{G}}_\theta \quad (82)$$

$$\iff \epsilon_\theta - \epsilon_{\text{target}} = \mathbf{x}_t + \alpha_t \hat{\mathbf{G}}_\theta - (\mathbf{x}_t + \alpha_t \hat{\mathbf{G}}_{\text{target}}) \quad (83)$$

$$\iff \epsilon_\theta - \epsilon_{\text{target}} = \alpha_t (\hat{\mathbf{G}}_\theta - \hat{\mathbf{G}}_{\text{target}}) \quad (84)$$

In both cases, $(\hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) - \hat{\mathbf{G}}_{\text{target}})$ can be rewritten to $(\epsilon_\theta(\mathbf{x}_t, s, t) - \epsilon_{\text{target}})$ by multiplying a factor α_t , and the guidance vector now matches that of the ELBO-diffusion loss. Therefore, we are motivated to incorporate α_t into $w(s, t)$ as proposed.

The exponent a for α_t^a takes a value of either 1 or 2. We explain the reason for each decision here. When $a = 1$, we guide the gradient $\frac{\partial}{\partial \theta} \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$, with score difference $(\epsilon_\theta(\mathbf{x}_t, s, t) - \epsilon_{\text{target}})$. To motivate $a = 2$, we first note that the weighted gradient

$$\tilde{w}(s, t) \frac{\partial}{\partial \theta} \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) = \frac{1}{|c_{\text{out}}(s, t)|} \frac{\partial}{\partial \theta} \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) = \pm \frac{\partial}{\partial \theta} \hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t)$$

and as shown above that ϵ -prediction is parameterized as

$$\epsilon_\theta(\mathbf{x}_t, s, t) = \mathbf{x}_t + \alpha_t \hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t).$$

Multiplying an additional α_t to $\frac{\partial}{\partial \theta} \hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t)$ therefore implicitly reparameterizes our model into an ϵ -prediction model. The case of $a = 2$ therefore implicitly reparameterizes our model into an ϵ -prediction model guided by the score difference $(\epsilon_\theta(\mathbf{x}_t, s, t) - \epsilon_{\text{target}})$. Empirically, α_t^2 additionally downweights loss for larger t compared to α_t , allowing the model to train on smaller time-steps more effectively.

Lastly, the division of $\alpha_t^2 + \sigma_t^2$ is inspired by the increased weighting for middle time-steps (Esser et al., 2024) for Flow Matching training. This is purely an empirical decision.

D. Training Algorithm

We present the training algorithm in Algorithm 3.

E. Classifier-Free Guidance

We refer readers to Appendix C.5 for analysis of each parameterization. Most notably, the network $\mathbf{G}_\theta$ in both (1) Simple-EDM with cosine diffusion schedule and (2) Euler-FM with OT-FM schedule are equivalent to v -prediction parameterization

Algorithm 3 IMM Training

Input: model $\mathbf{f}^\theta$, data distribution $q(\mathbf{x})$ and label distribution $q(\mathbf{c}|\mathbf{x})$ (if label is used), prior distribution $\mathcal{N}(0, \sigma_d^2 I)$, time distribution $p(t)$ and $p(s|t)$, DDIM interpolator $\text{DDIM}(\mathbf{x}_t, \mathbf{x}, s, t)$ and its flow coefficients α_t, σ_t , mapping function $r(s, t)$, kernel function $k(\cdot, \cdot)$, weighting function $w(s, t)$, batch size B , particle number M , label dropout probability p

Output: learned model $\mathbf{f}^\theta$

Initialize $n \leftarrow 0, \theta^0 \leftarrow \theta$

while model not converged **do**

Sample a batch of data, label, and prior, and split into B/M groups, $\{(x^{(i,j)}, c^{(i,j)}, \epsilon^{(i,j)})\}_{i=1, j=1}^{B/M, M}$

For each group, sample $\{(s^i, t^i)\}_{i=1}^{B/M}$ and $r^i = r(s^i, t^i)$ for each i . This results in a tuple $\{(s^i, r^i, t^i)\}_{i=1}^{B/M}$

$x_{t^i}^{(i,j)} \leftarrow \text{DDIM}(\epsilon^{(i,j)}, x^{(i,j)}, t^i, 1) = \alpha_{t^i} x^{(i,j)} + \sigma_{t^i} \epsilon^{(i,j)}, \forall (i, j)$

$x_{r^i}^{(i,j)} \leftarrow \text{DDIM}(x_{t^i}^{(i,j)}, x^{(i,j)}, r^i, t^i), \forall (i, j)$

(Optional) Randomly drop each label $c^{(i,j)}$ to be null token $\emptyset$ with probability p

$\theta_{n+1} \leftarrow$ optimizer step by minimizing $\hat{\mathcal{L}}_{\text{IMM}}(\theta_n)$ using model $\mathbf{f}^{\theta_n}$ (see Eq. (67)) (optionally inputting $c^{(i,j)}$ into network)

end while

in diffusion (Salimans & Ho, 2022) and FM (Lipman et al., 2022). When conditioned on label $\mathbf{c}$ during sampling, it is customary to use classifier-free guidance to reweight this v -prediction network via

$$\begin{aligned} & \mathbf{G}_\theta^w(c_{\text{in}}(t)\mathbf{x}_t, c_{\text{noise}}(s), c_{\text{noise}}(t), \mathbf{c}) \\ &= w \mathbf{G}_\theta(c_{\text{in}}(t)\mathbf{x}_t, c_{\text{noise}}(s), c_{\text{noise}}(t), \mathbf{c}) + (1 - w) \mathbf{G}_\theta(c_{\text{in}}(t)\mathbf{x}_t, c_{\text{noise}}(s), c_{\text{noise}}(t), \emptyset) \end{aligned} \quad (85)$$

with guidance weight w so that the classifier-free guided $\mathbf{f}_{s,t,w}^\theta(\mathbf{x}_t)$ is

$$\mathbf{f}_{s,t,w}^\theta(\mathbf{x}_t) = c_{\text{skip}}(s, t)\mathbf{x}_t + c_{\text{out}}(s, t) \mathbf{G}_\theta^w(c_{\text{in}}(t)\mathbf{x}_t, c_{\text{noise}}(s), c_{\text{noise}}(t), \mathbf{c}) \quad (86)$$

F. Sampling Algorithms

Pushforward sampling. See Algorithm 4. We assume a series of N time steps $\{t_i\}_{i=0}^N$ with $T = t_N > t_{N-1} > \dots > t_2 > t_1 > t_0 = \epsilon$ for the maximum time T and minimum time ϵ . Denote σ_d as data standard deviation.

Restart sampling. See Algorithm 5. Different from pushforward sampling, N time steps $\{t_i\}_{i=0}^N$ do not need to be strictly decreasing for all time steps, e.g. $T = t_N \geq t_{N-1} \geq \dots \geq t_2 \geq t_1 \geq t_0 = \epsilon$ (assuming $T > \epsilon$). Different from pushforward sampling, restart sampling first denoise a clean sample before resampling a noise to be added to this clean sample. Then a clean sample is predicted again. The process is iterated for N steps.

G. Connection with Prior Works

G.1. Consistency Models

Consistency models explicitly match PF-ODE trajectories using a network $\mathbf{g}_\theta(\mathbf{x}_t, t)$ that directly outputs a sample given any $\mathbf{x}_t \sim q_t(\mathbf{x}_t)$. The network explicitly uses EDM parameterization to satisfy boundary condition $\mathbf{g}_\theta(\mathbf{x}_0, 0) = \mathbf{x}_0$ and trains via loss $\mathbb{E}_{\mathbf{x}_t, \mathbf{x}_r, t} [\|\mathbf{g}_\theta(\mathbf{x}_t, t) - \mathbf{g}_\theta(\mathbf{x}_r, r)\|^2]$ where $\mathbf{x}_r$ is a deterministic function of $\mathbf{x}_t$ from an ODE solver.

We show that CM loss is a special case of our simplified IMM objective.

Lemma 1. When $\mathbf{x}_t = \mathbf{x}'_t$, $\mathbf{x}_r = \mathbf{x}'_r$, $k(x, y) = -\|x - y\|^2$, and $s > 0$ is a small constant, Eq. (12) reduces to CM loss $\mathbb{E}_{\mathbf{x}_t, \mathbf{x}_r, t} [w(t) \|\mathbf{g}_\theta(\mathbf{x}_t, t) - \mathbf{g}_\theta(\mathbf{x}_r, r)\|^2]$ for some valid mapping $r(t) < t$.

Proof. Since $\mathbf{x}_t = \mathbf{x}'_t$, $\mathbf{x}_r = \mathbf{x}'_r$, we have $\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) = \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)$ and $\mathbf{f}_{s,r}^\theta(\mathbf{x}_r) = \mathbf{f}_{s,r}^\theta(\mathbf{x}'_r)$. So $k(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)) = k(\mathbf{f}_{s,r}^\theta(\mathbf{x}_r), \mathbf{f}_{s,r}^\theta(\mathbf{x}'_r)) = 0$ by definition. Since $k(x, y) = -\|x - y\|^2$, it is easy to see Eq. (12) reduces to

$$\mathbb{E}_{\mathbf{x}_t, t} [w(s, t) \|\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^\theta(\mathbf{x}_r)\|^2] \quad (87)$$